

Satellite Structural Inspection Interferometric Synthetic Aperture Radar (InSAR) Based

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METHODOLOGY

1. Satellite SAR System with Resolution and Revisit Time

To select a suitable SAR system, technical specifications such as spatial resolution, frequency band, polarization, and revisit time were considered. Sentinel-1, operating at C-band (5.405 GHz), was selected due to:

Parameter	Sentinel-1 C-SAR (IW Mode)
Frequency Band	C-band (~5.405 GHz)
Revisit Time	12 days (single satellite)
Spatial Resolution	5 m (range) × 20 m (azimuth)
Coverage Mode	Interferometric Wide Swath (IW)

2. Retrieving Multi-Temporal SAR Imagery Over the Specified Period

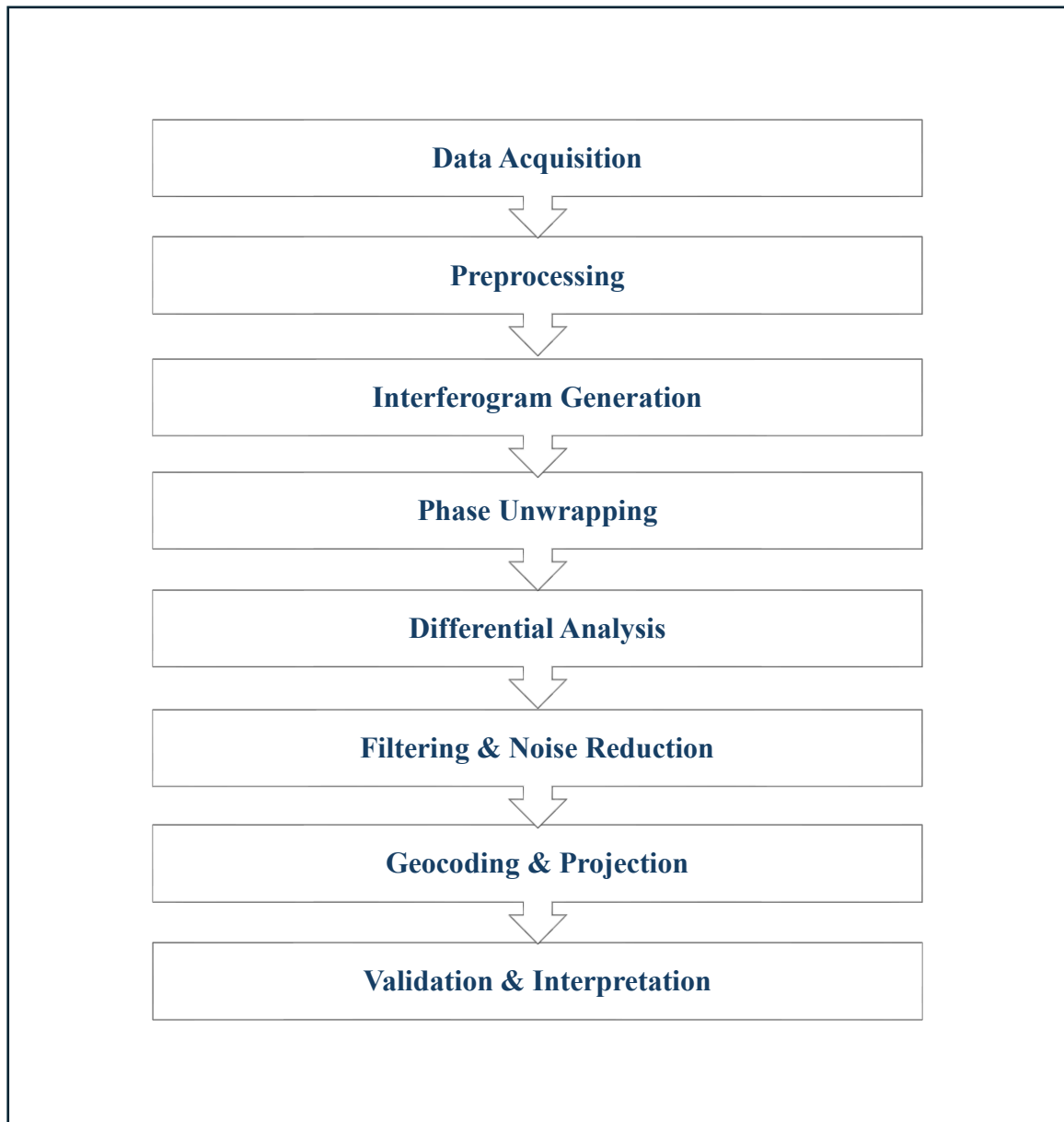
Multi-temporal Sentinel-1 SLC IW mode data were downloaded based on:

- **Area of Interest (AOI):** A polygon covering the bridge and surrounding terrain.
- **Date range:** Multiple acquisitions were collected spanning several months to capture temporal variations.
- **Image acquisition conditions:** All images were selected with the same relative orbit, pass direction (ascending or descending), and polarization (typically VV/VH) to ensure consistency for interferometric analysis.

S1A_IW_SLC__1SDV_20250111T130240_20250111T130310_057397_0710AB_813D
S1A_IW_SLC__1SDV_20250123T130239_20250123T130309_057572_07179E_5431
S1A_IW_SLC__1SDV_20250204T130238_20250204T130308_057747_071E86_3FE6
S1A_IW_SLC__1SDV_20250216T130238_20250216T130308_057922_072592_C312
S1A_IW_SLC__1SDV_20250228T130238_20250228T130307_058097_072CB4_7A14
S1A_IW_SLC__1SDV_20250312T130238_20250312T130308_058272_0733CA_217F
S1A_IW_SLC__1SDV_20250324T130238_20250324T130308_058447_073AB5_B445
S1A_IW_SLC__1SDV_20250405T130238_20250405T130308_058622_0741CE_2B52
S1A_IW_SLC__1SDV_20250417T130238_20250417T130308_058797_0748F4_3FFD
S1A_IW_SLC__1SDV_20250429T130239_20250429T130309_058972_07501E_08D8
S1A_IW_SLC__1SDV_20250511T130238_20250511T130308_059147_0756BA_DE13
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S1A_IW_SLC__1SDV_20250628T130236_20250628T130306_059847_076EE3_6BFD

3. Differential SAR Interferometry (D-InSAR) for Displacement Measurement

D-InSAR analysis was performed to measure millimeter-scale deformation over time:



VISUALIZATIONS

1. Defining specific bridge location & parameters for satellite SAR based inspection



Project & Bridge Details

State – Rajasthan

Bridge No. – 970/1

Bridge Chainage – 968+730

Project - Chittorgarh to Kota

Zone – RJ 06

Structure Type – MNB

Sar Parameters

Sensor – Sentinel-1 (C-Band)

Beam Mode – IW (Interferometric Wide Swath)

Path – 34

Frame – 519

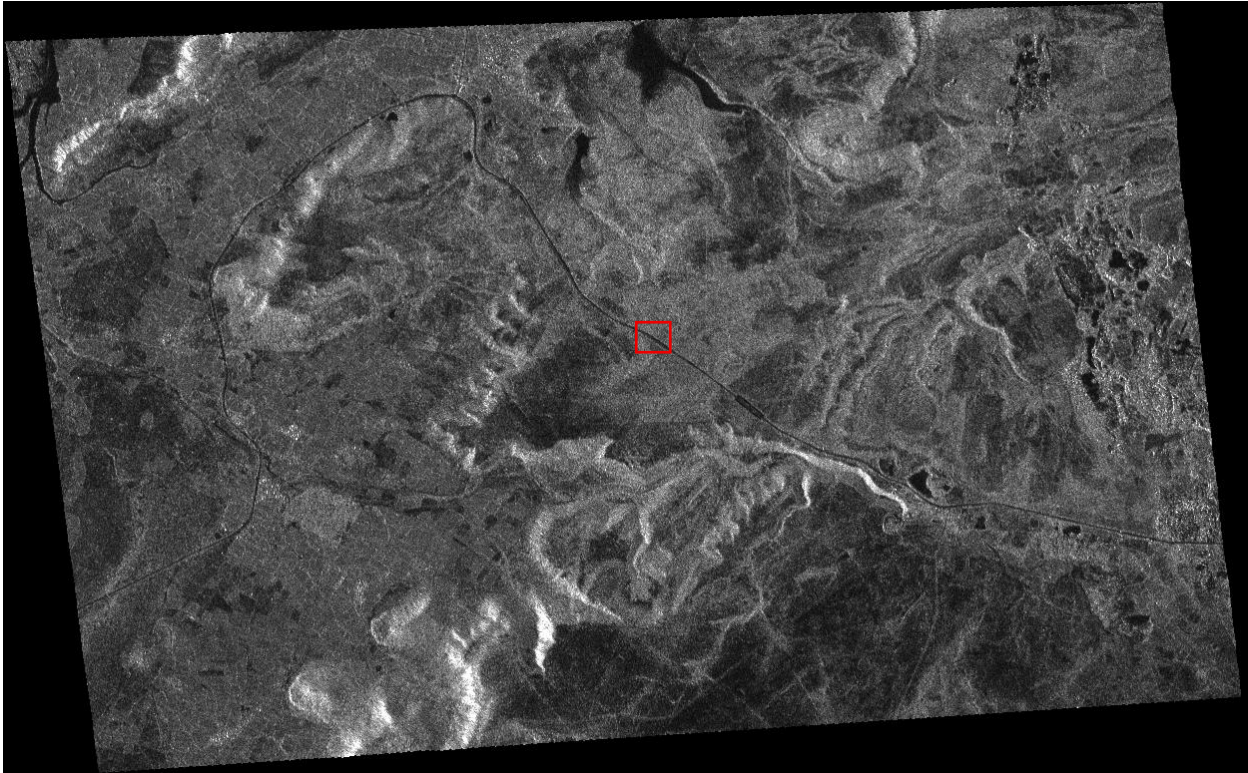
Flight Direction - DESCENDING

Polarization – VV+VH

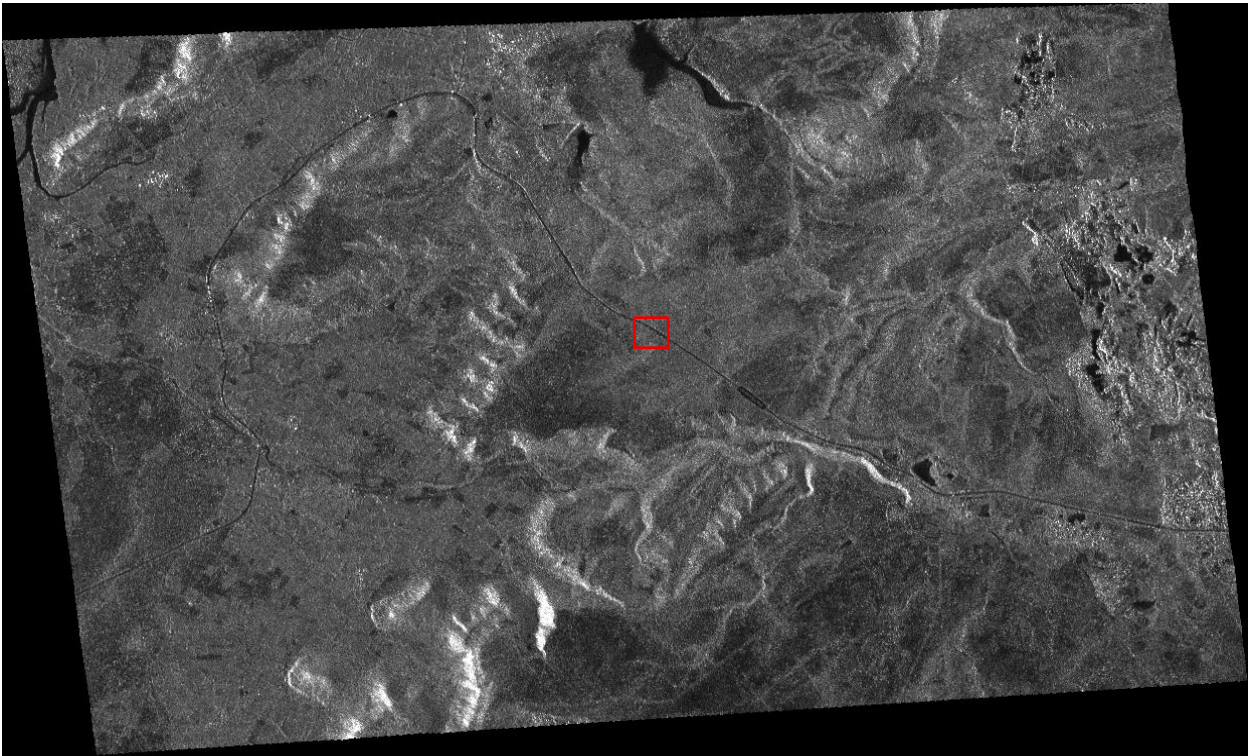
Absolute Orbit – 54706

2. Calibration, georeferencing, orthorectification of SAR images (VH/VV)

- Orthorectification of SAR image VH Polarization



- Orthorectification of SAR image VV Polarization



3. Results of SAR based bridge inspection, including visualizations

Bridge Components

MNB – CH-968+730

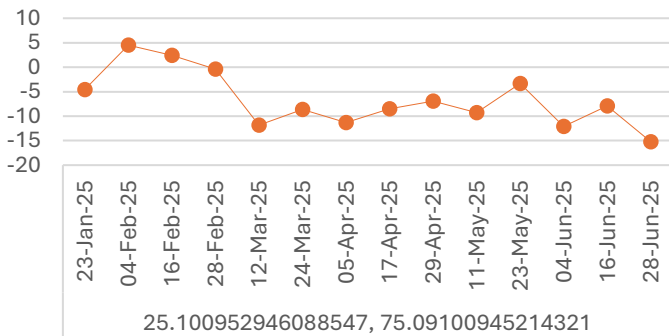
Length – 900M



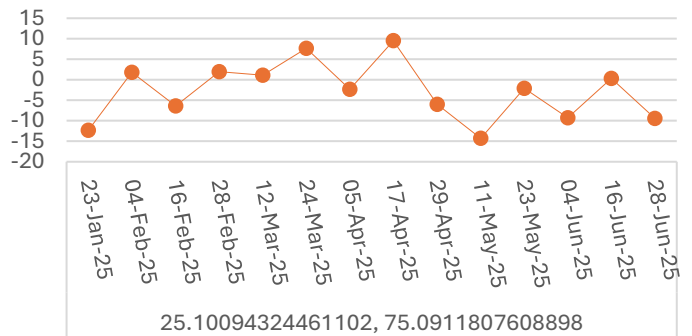
Scan Duration		From : 23 Jan 2025	To : 28 June 2025
Components & Abbreviation			
A1		Abutment 1	
P1		Pier 1	
P2		Pier 2	
A2		Abutment 2	
LHS		Left Hand Side	
RHS		Right Hand Side	

3.1 VV Displacement From 23 Jan 2025 To 28 June 2025

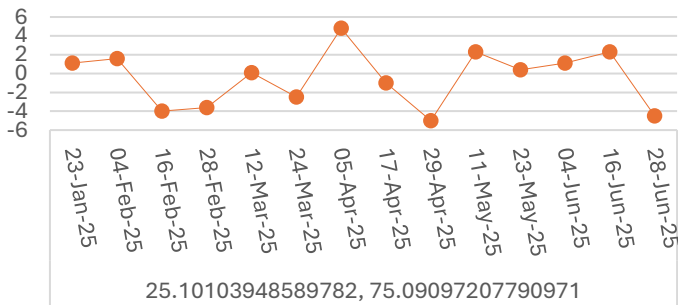
Displacement @ A1 LHS VV



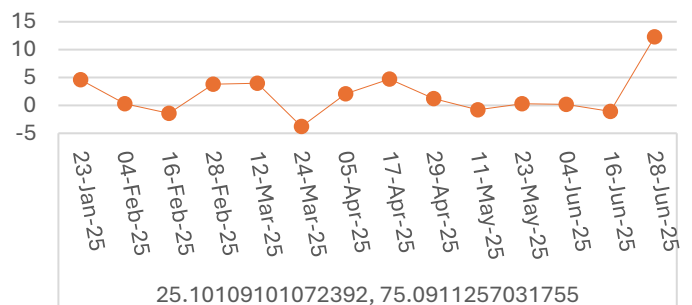
Displacement @ A1 RHS VV



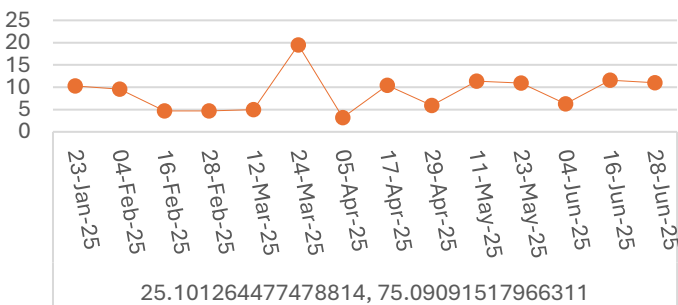
Displacement @P1 LHS VV



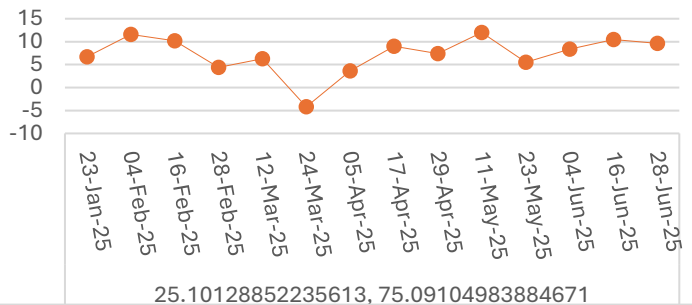
Displacement @P1 RHS VV



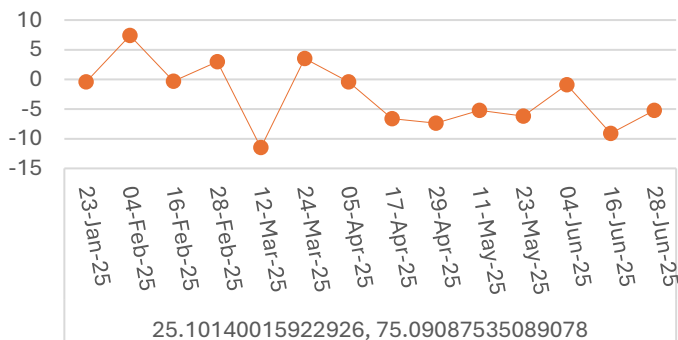
Displacement @P2 LHS VV



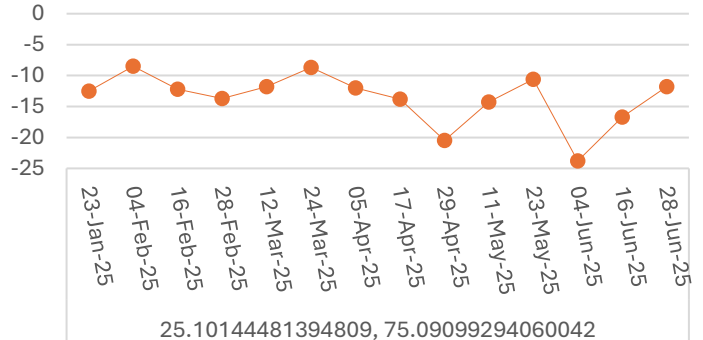
Displacement @P2 RHS VV



Displacement @A2 LHS VV

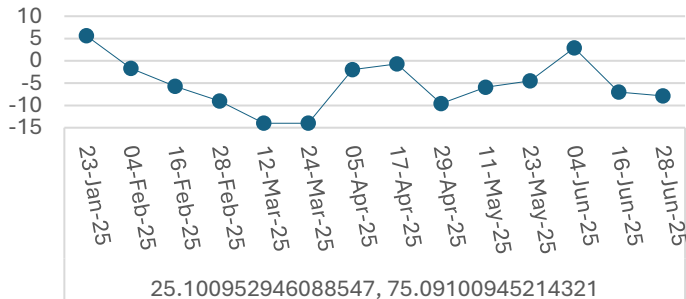


Displacement @A2 RHS VV

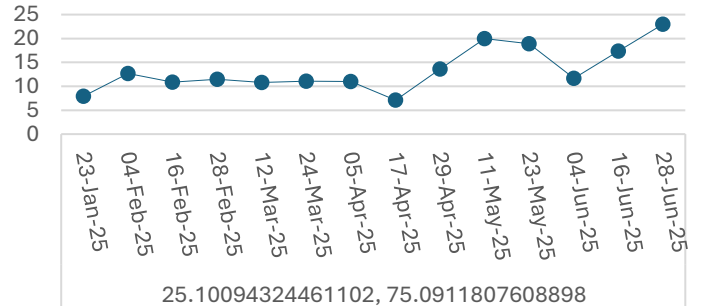


3.2 VH Displacement From 23 Jan 2025 To 28 June 2025

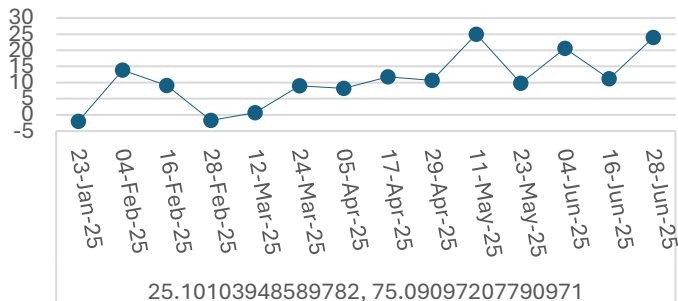
Displacement @A1 LHS VH



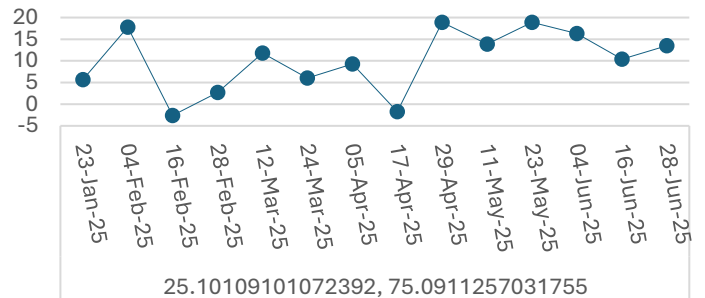
Displacement @A1 RHS VH



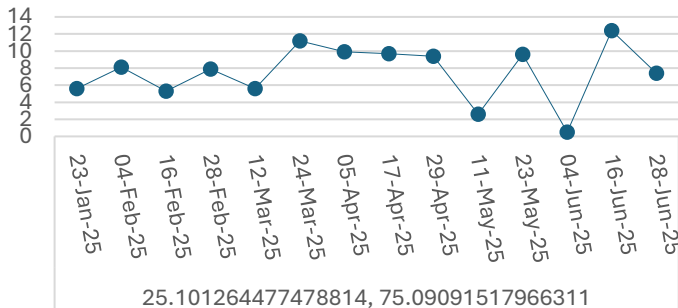
Displacement @P1 LHS VH



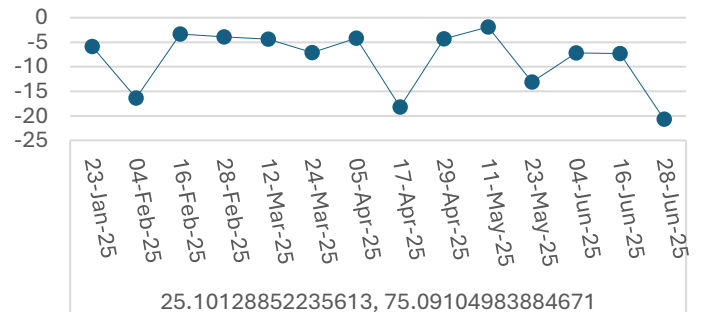
Displacement @P1 RHS VH



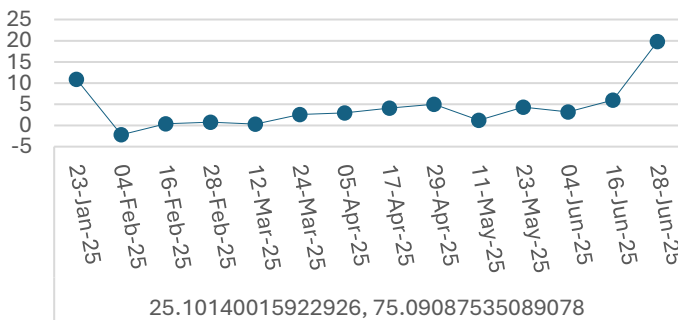
Displacement @P2 LHS VH



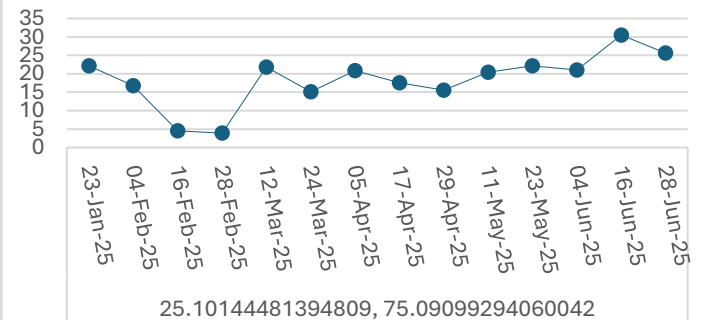
Displacement @P2 RHS VH



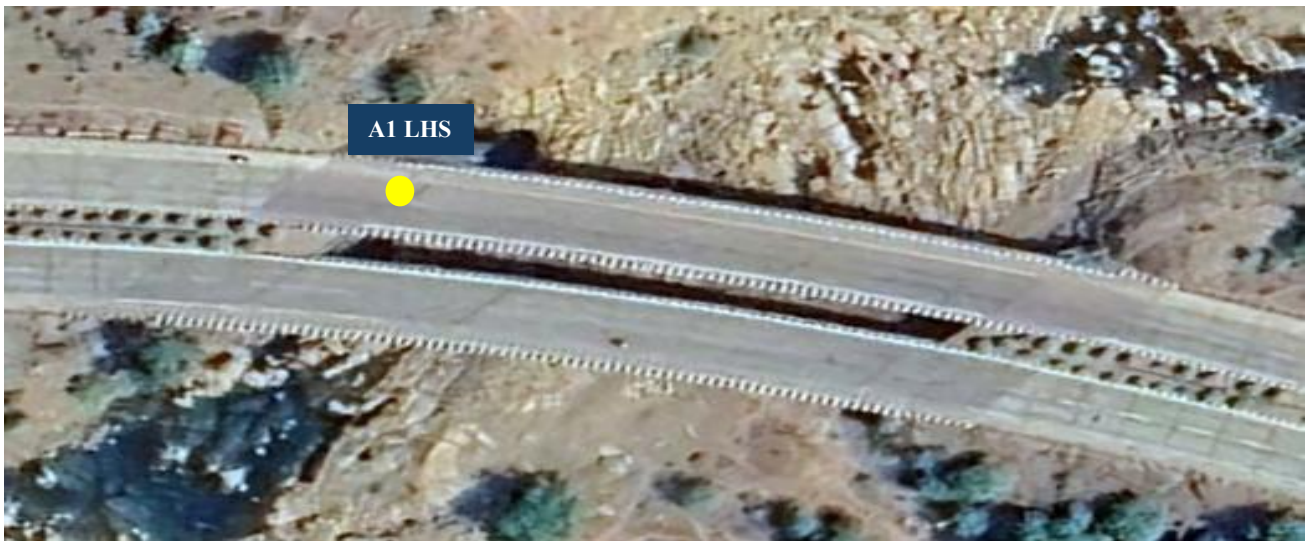
Displacement @A2 LHS VH



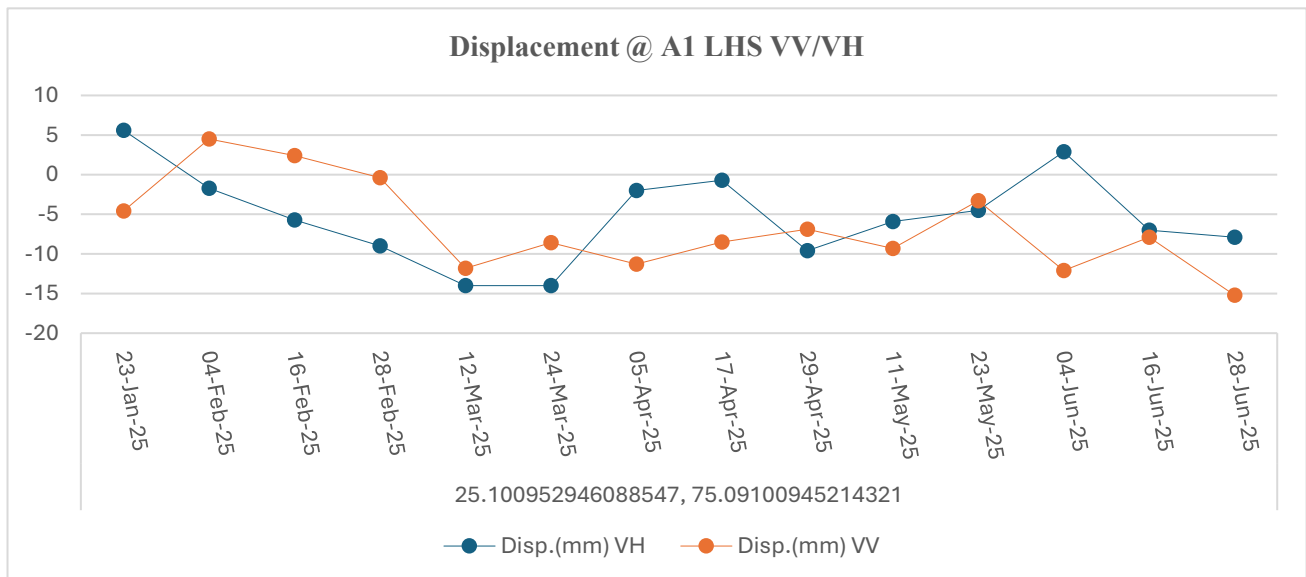
Displacement @A2 RHS VH



3.3 SAR Based Analysis at Abutment 1 LHS:



A1 LHS (25.100952946088547, 75.09100945214321)



Key Observations:

Displacement Magnitude Patterns:

- **Vertical Displacement (VV)** Vertical movements exhibit moderate variability, generally ranging from approximately -15 mm to +5 mm. The trend shows alternating uplift and settlement phases, consistent with thermal effects and variable service loading rather than sustained deformation
- **Horizontal displacement (VH)** Horizontal displacements remain comparatively restrained, mostly within -15 mm to +5 mm. The overall pattern is oscillatory with no persistent unidirectional drift, indicating stable lateral behaviour of the monitored point

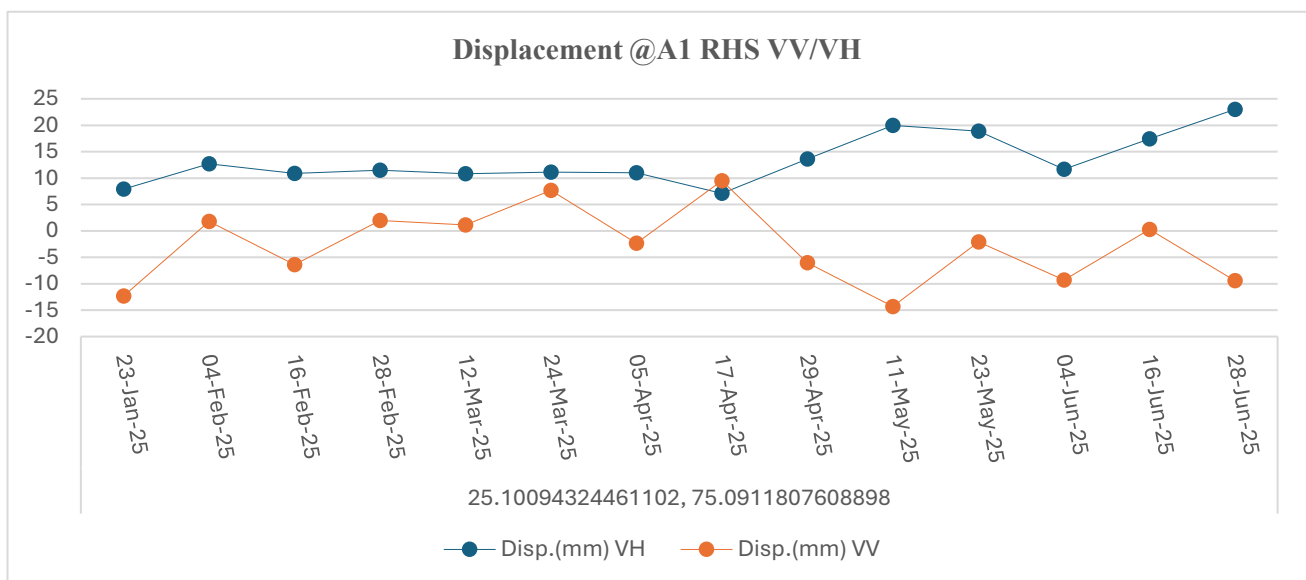
Critical Events & Anomalies:

- **March 2025:** Both VH and VV reach pronounced negative values (around -14 to -16 mm), representing the most significant combined movement during the monitoring period.
- **June 2025:** VV again trends downward toward -15 mm, while VH stabilizes near -7 to -8 mm, showing no accelerating or cumulative displacement pattern.

3.4 SAR Based Analysis at Abutment 1 RHS:



A1 RHS (25.10094324461102, 75.0911807608898)



Key Observations:

Displacement Magnitude Patterns:

- **Horizontal displacement (VH)** Horizontal displacements remain consistently positive and restrained, generally varying between +8 mm and +24 mm.
- **Vertical displacement (VV)** Vertical movements show moderate variability, predominantly ranging from approximately -18 mm to +5 mm.

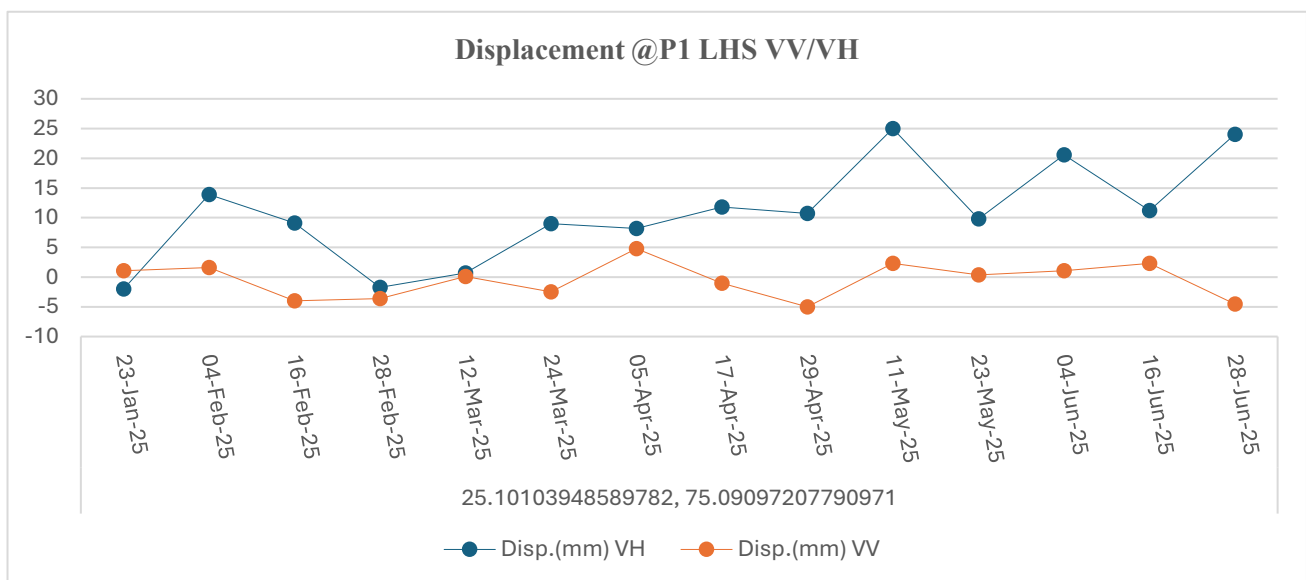
Critical Events & Anomalies:

- **April-May 2025:** VH increases noticeably, reaching values above 20 mm, marking the peak horizontal response during the monitoring window. This increase appears gradual and uniform, not sudden, indicating controlled lateral behaviour.
- **May 2025:** VV records its most significant downward excursion (approximately -18 mm), representing the maximum vertical response observed. The absence of a corresponding sharp VH anomaly suggests localized vertical response rather than global structural distress.

3.5 SAR Based Analysis at Pier 1 LHS:



P1 LHS (25.10103948589782, 75.09097207790971)



Key Observations:

Displacement Magnitude Patterns:

- **Vertical Displacement (VV):** Vertical movements show mild to moderate variability, generally ranging between approximately -6 mm and $+5$ mm. The pattern is characterized by alternating uplift and settlement phases without any sustained downward or upward trend.
- **Horizontal displacements (VH)** Horizontal displacements exhibit a **more pronounced range**, varying roughly from -3 mm to $+25$ mm. While VH shows noticeable oscillations and several peaks, the overall behaviour reflects gradual lateral movement rather than abrupt shifts.

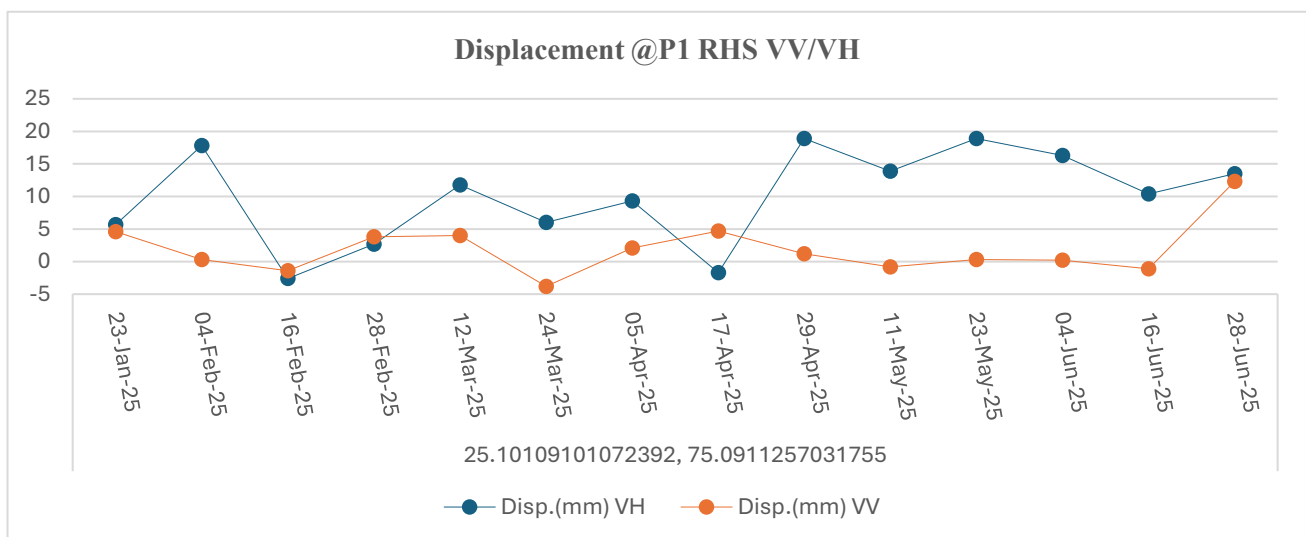
Critical Events & Anomalies:

- **May 2025:** A clear peak in horizontal displacement is observed (around $+25$ mm), representing the maximum VH value in the monitoring period. This appears as an isolated event rather than a continuously escalating trend
- **June-July 2025:** VH remains relatively elevated (around $+20$ to $+24$ mm) with VV remaining near neutral to slightly negative values, indicating lateral movement dominance without corresponding vertical distress.

3.6 SAR Based Analysis at Pier 1 RHS:



P1 RHS (25.10109101072392, 75.0911257031755)



Key Observations:

Displacement Magnitude Patterns:

- **Vertical displacement (VV):** Vertical displacement values show low to moderate variability, generally fluctuating between approximately -6 mm and +6 mm.
- **Horizontal displacement(VH):** Horizontal displacement exhibits greater variability and higher magnitudes compared to VV, ranging roughly from -3 mm to +20 mm. The pattern shows recurrent oscillations with multiple peaks, suggesting progressive lateral movement rather than sudden or unstable shifts.

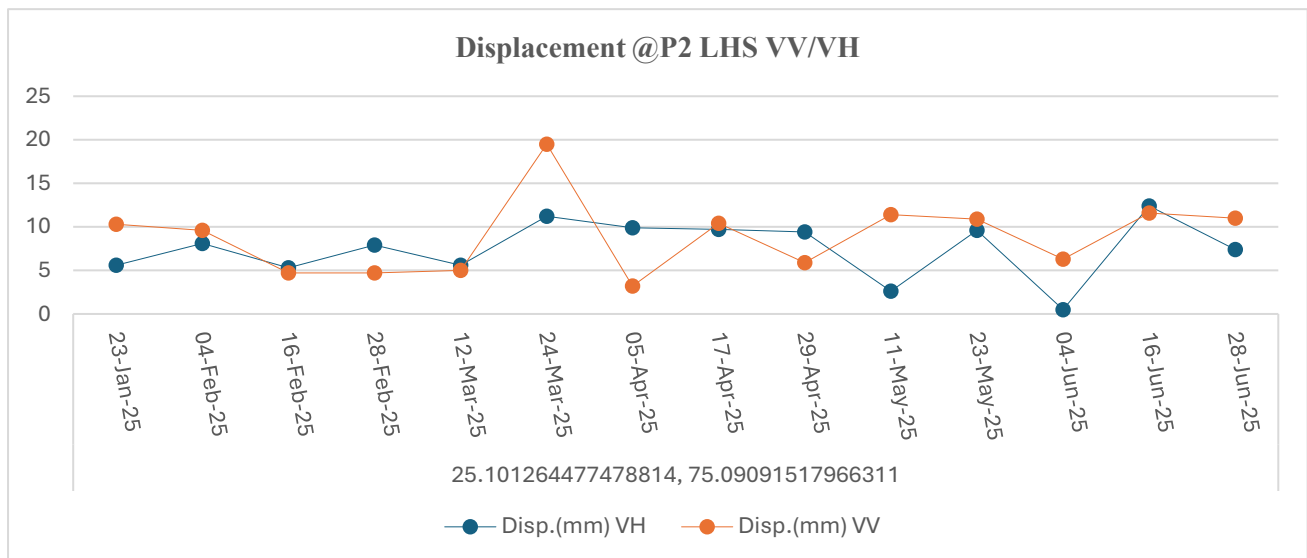
Critical Events & Anomalies:

- **April-May 2025:** A pronounced peak in VH displacement is observed, reaching approximately +18 to +20 mm, marking the maximum horizontal movement recorded in the dataset. This appears as a localized peak, not part of a continuous acceleration trend.
- **June 2025:** VH displacement remains moderately elevated (around +12 to +15 mm) while VV values stay close to neutral or slightly negative, implying dominant lateral movement without corresponding vertical instability.

3.7 SAR Based Analysis at Pier 2 LHS:



P2 LHS (25.101264477478814, 75.09091517966311)



Key Observations:

Displacement Magnitude Patterns:

- **Vertical displacement (VV):** Vertical displacement shows moderate variability, generally ranging between approximately +3 mm and +20 mm. The pattern is characterized by short-term fluctuations with intermittent uplift peaks, rather than a consistent upward or downward trend.
- **Horizontal displacement(VH):** Horizontal displacement values fluctuate within a narrower band compared to VV, typically ranging from approximately 0 mm to +12 mm.

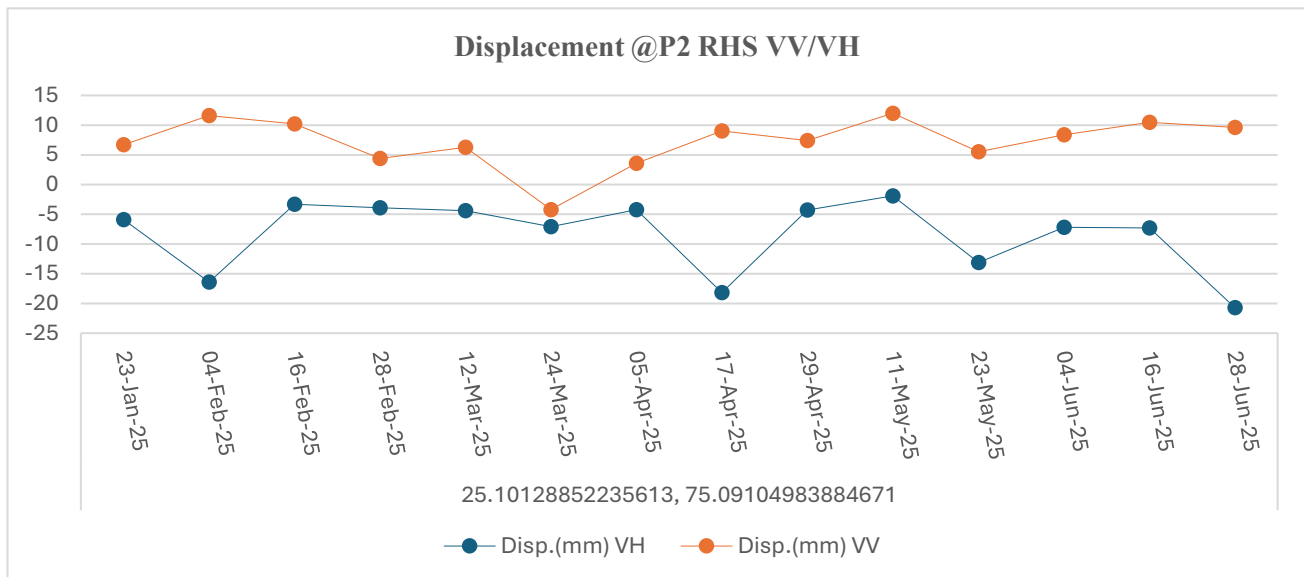
Critical Events & Anomalies:

- **March 2025:** A distinct peak in vertical displacement is observed, reaching approximately +20 mm, representing the maximum VV value during the monitoring period. This appears as an isolated uplift event, not followed by continued escalation.
- **June 2025:** Both VH and VV values rise concurrently (VH +12 mm, VV +11 mm), suggesting a short period of combined lateral and vertical response, though still within moderate limits.

3.8 SAR Based Analysis at Pier 2 RHS:



P2 RHS (25.10128852235613, 75.09104983884671)



Key Observations:

Displacement Magnitude Patterns:

- **Vertical displacement (VV):** Vertical displacement values show moderate variability, generally ranging between approximately +3 mm and +12 mm.
- **Horizontal displacement (VH)** Horizontal displacement exhibits more pronounced variability and remains consistently **negative**, ranging approximately from –5 mm to –22 mm.

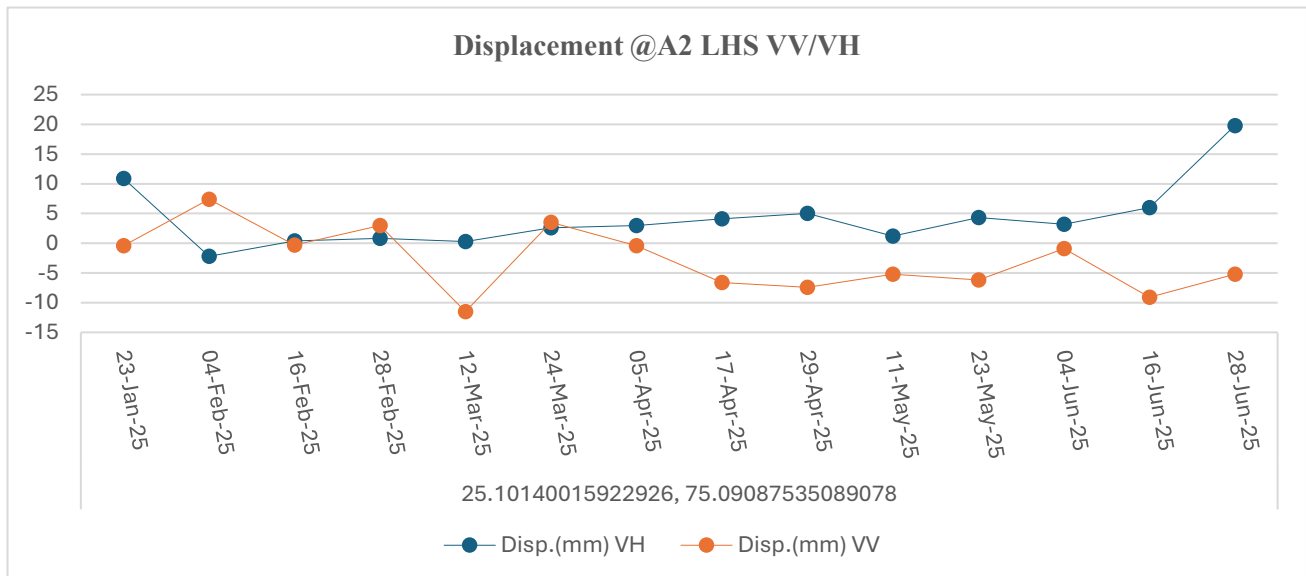
Critical Events & Anomalies:

- **February 2025:** A notable negative peak in VH displacement is observed (around –17 mm), representing one of the larger lateral movement events during the monitoring period.
- **April 2025:** VH reaches its maximum negative displacement, marking the most significant horizontal movement recorded at this location.
- **June 2025:** VH displacement again declines sharply (around –22 mm), indicating a renewed lateral movement episode, while VV remains stable and positive, suggesting horizontal movement dominance without vertical distress.

3.9 SAR Based Analysis at Abutment 2 LHS:



A2 LHS (25.10140015922926, 75.09087535089078)



Key Observations:

Displacement Magnitude Patterns:

- **Vertical displacement (VV):** Vertical displacement demonstrates moderate variability, generally fluctuating between approximately -12 mm and +8 mm.
- **Horizontal displacement (VH):** Horizontal displacement remains comparatively restrained for most of the monitoring period, typically ranging between -2 mm and +6 mm.

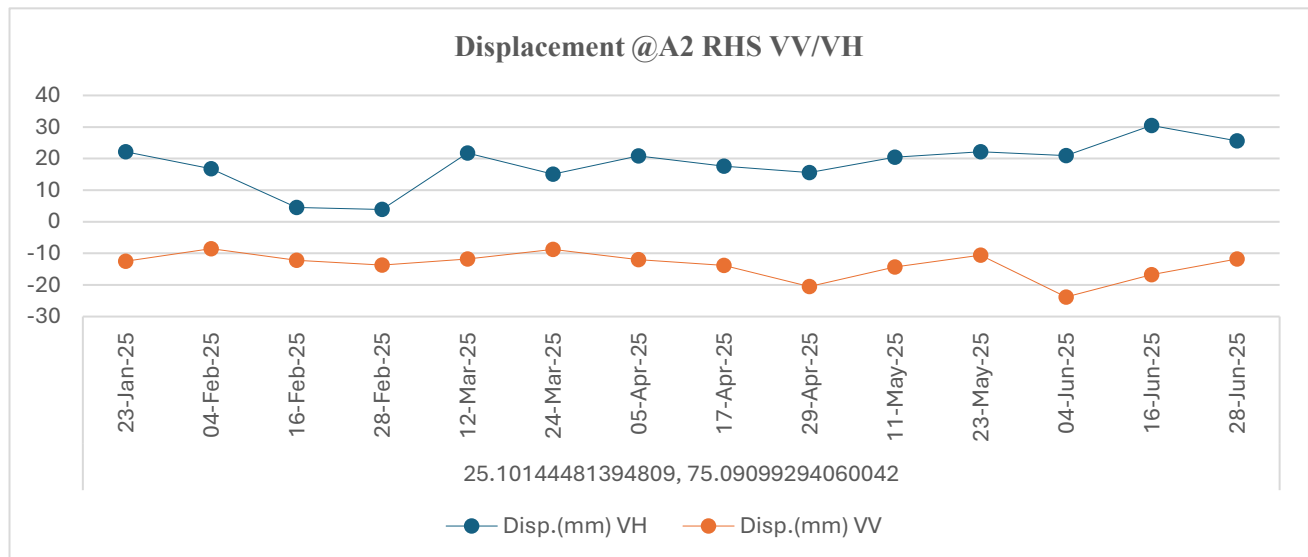
Critical Events & Anomalies:

- **February 2025:** Both VV and VH show short-term fluctuations, but movements remain within moderate limits, indicating stable structural response without any abrupt deformation.
- **April –May 2025:** VH exhibits minor oscillations with no persistent directional drift, while VV stabilises closer to neutral, suggesting temporary load- or temperature-driven movements rather than structural distress.
- **June 2025:** VH increases sharply to approximately +20 mm, marking the maximum horizontal displacement recorded at this location.

3.10 SAR Based Analysis at Abutment 2 LHS:



A2 RHS (25.10144481394809, 75.09099294060042)



Key Observations:

Displacement Magnitude Patterns:

- **Vertical displacement (VV):** Vertical displacement values remain predominantly negative throughout the monitoring period, generally ranging between approximately –
- **Horizontal displacement (VH)** Horizontal displacement exhibits moderate to high variability, with values ranging approximately from +5 mm to +30 mm.

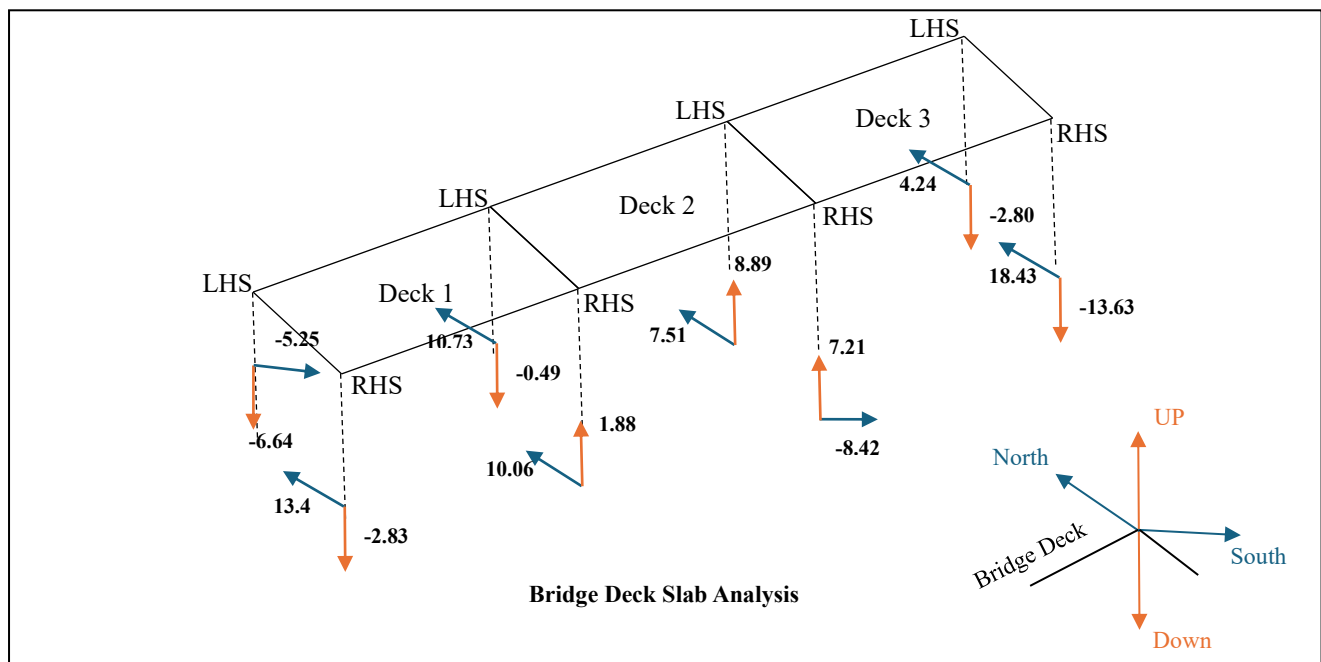
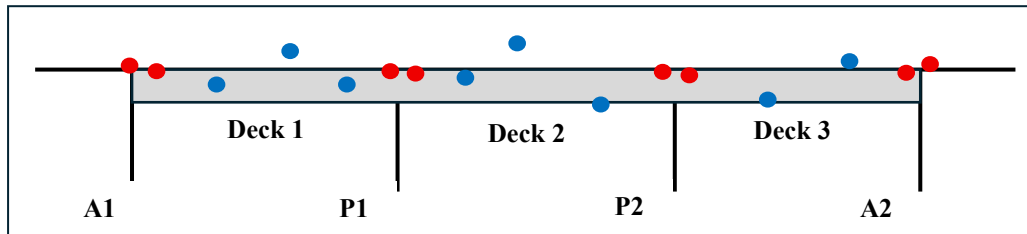
Critical Events & Anomalies:

- **January-February 2025:** VH shows an initial decline from around +22 mm to +5 mm, indicating a short-term reduction in lateral displacement.
- **April 2025:** VV reaches one of its maximum negative values (–25 mm), marking the most significant vertical settlement recorded at this location.
- **June 2025:** VH peaks at approximately +30 mm, representing the maximum horizontal displacement observed over the monitoring period.

ANALYSIS

1. Identify potential deformation, shifts or movement of the bridge structure

- Points of Analysis (POA) of bridge (●)



Vertical Displacement Behaviour:

- The vertical displacements observed across Decks 1, 2, and 3 show a combination of modest upward and downward movements, generally within a limited range (approximately -14 mm to +9 mm). These values are typical of bridge deck slabs responding to thermal gradients, daily temperature cycles, and routine traffic loading.

Horizontal Displacement Behaviour:

- Horizontal (lateral) displacements across all three decks range from small to moderate magnitudes, generally within ± 20 mm, which is consistent with expected superstructure movements due to thermal expansion-contraction and minor deck rotations under live loads.

CONCLUSION:

- The combined vertical and horizontal displacement patterns for Decks 1, 2, and 3 are consistent with normal structural behaviour of a multi-span bridge deck slab subjected to environmental effects and service loads. All recorded movements remain within acceptable engineering limits, with no evidence of abnormal settlement, excessive uplift, or progressive lateral drift. Overall, the bridge deck slab system is performing satisfactorily at the serviceability level. Continued routine monitoring is recommended to confirm long-term trends and to ensure early detection of any future deviations from expected behaviour.